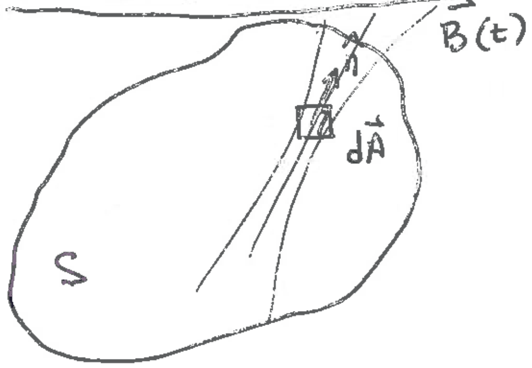
ELECTROMAGNETIC INDUCTION



$$\frac{d\Phi}{dt} = \frac{d}{dt}\int_S \vec{B}(t)\cdot d\vec{A}$$

$$= \int_S \frac{d\vec{B}(t)}{dt}\cdot d\vec{A} + \int_S \vec{B}(t)\cdot \cancel{\frac{d\vec{A}}{dt}}^{\,0}$$

$$\frac{d\Phi}{dt} = \int_S \frac{d\vec{B}(t)}{dt}\cdot d\vec{A}$$

FARADAY'S LAW

$$\oint_C \vec{E}\cdot d\vec{l} = -\frac{d}{dt}\underbrace{\int_S \vec{B}(t)\cdot d\vec{A}}$$

$$-\oint_C \vec{E}\cdot d\vec{l} = \frac{d\Phi}{dt} \leftarrow \text{CHANGING MAGNETIC FLUX}$$

LINE INTEGRAL

ELECTRIC FIELD

LORENTZ FORCE LAW

$$\vec{F} = q\left[\vec{E} + \cancel{\vec{v}}^{\,0}\times\vec{B}\right]$$

FORCE

ELECTRIC CHARGE

VELOCITY

$$\vec{F} = q\vec{E}$$

$\mathcal{E}$ = ELECTRO MOTIVE FORCE, EMF

$$\mathcal{E} = \oint_C \frac{\vec{F}\cdot d\vec{l}}{q} \qquad \frac{[N][m]}{[C]} = [V]$$

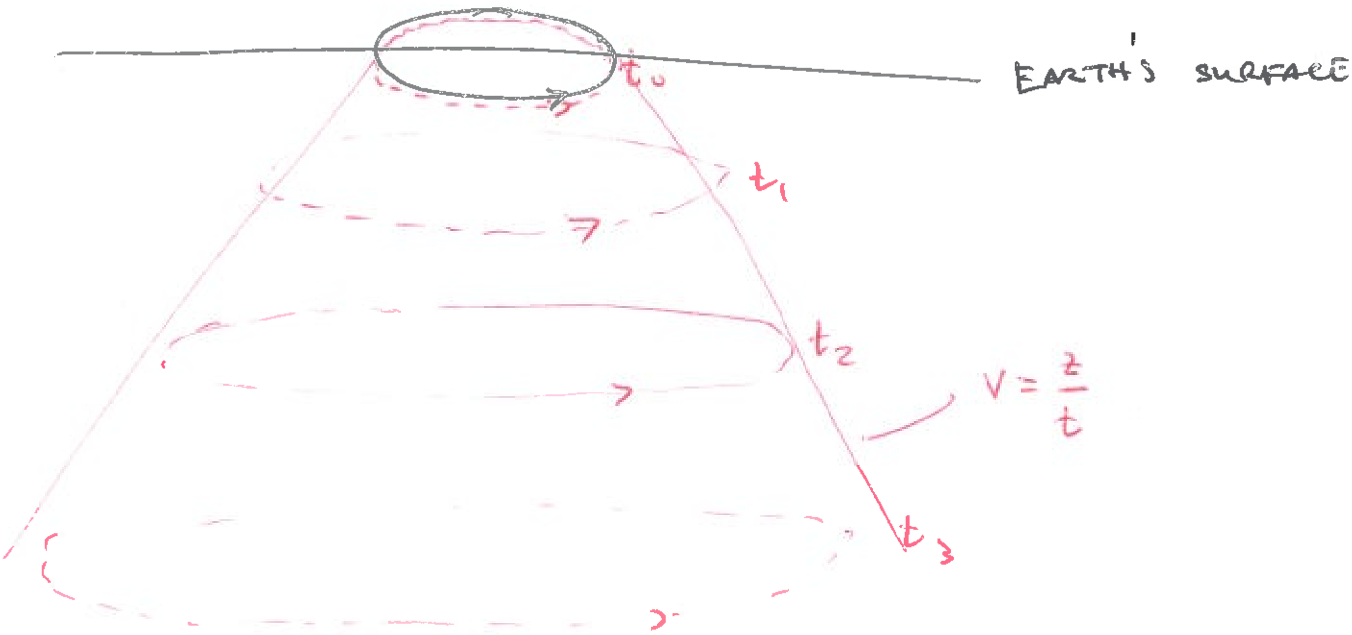
$$= \oint_C \vec{E}\cdot d\vec{l}$$

$$\mathcal{E} = -\frac{d\Phi}{dt}$$

$$\mathcal{E} = -\frac{d\Phi}{dt}$$

EMF

MAGNETIC FLUX CHANGING IN TIME



UNIFORM CONDUCTING MEDIUM

FIELD IS MAXIMUM @ DIFFUSION DEPTH

$$z = \sqrt{\frac{2t}{\sigma\mu}}$$

CONDUCTIVITY

MAGNETIC PERMEABILITY

TYPICALLY ASSUME $\mu = \mu_0$

$$z = vt$$

$$v = \sqrt{\frac{1}{2\sigma\mu t}}$$

MAXIMUM VELOCITY

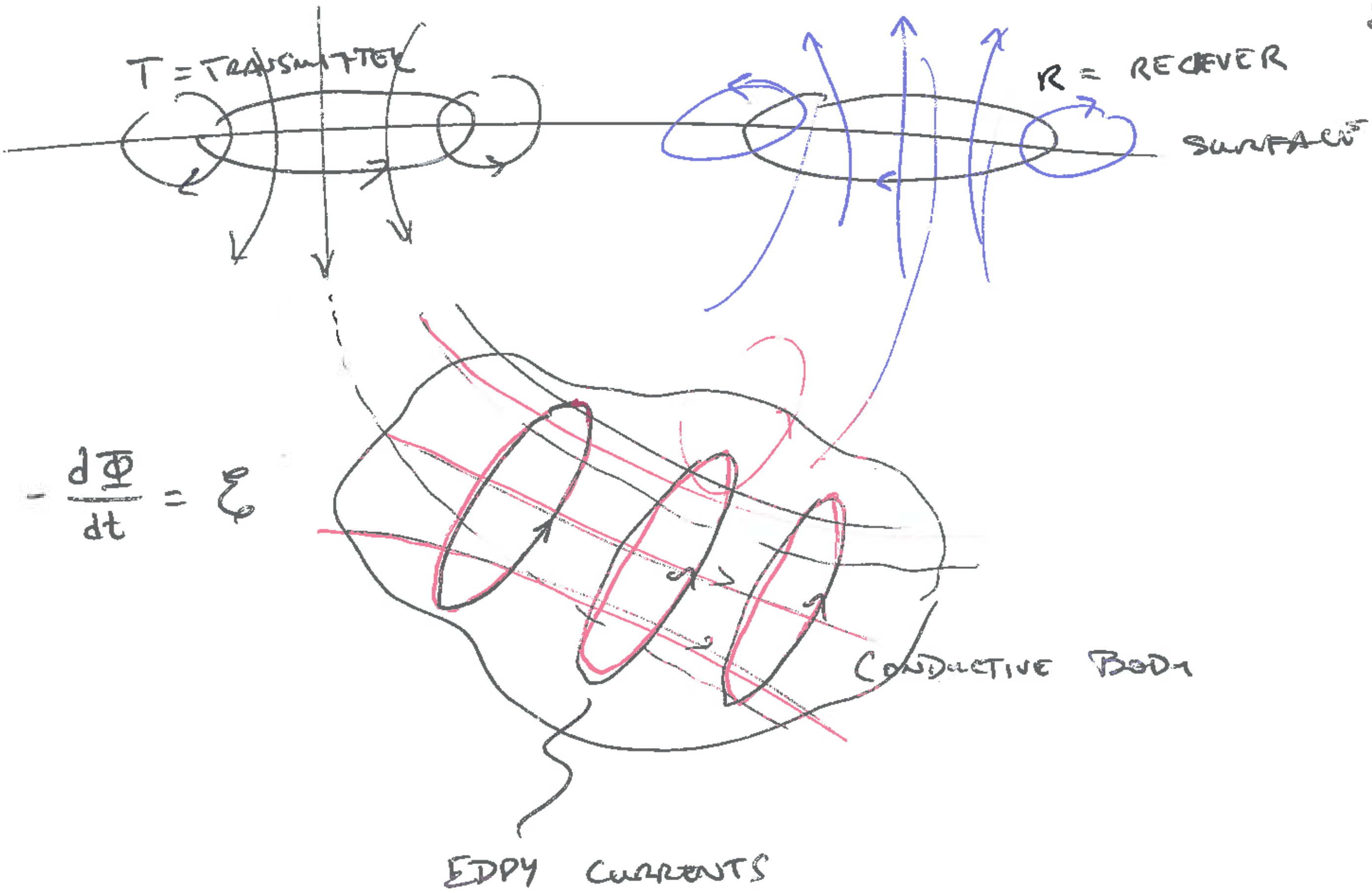
CONDUCTIVITY RESISTIVITY

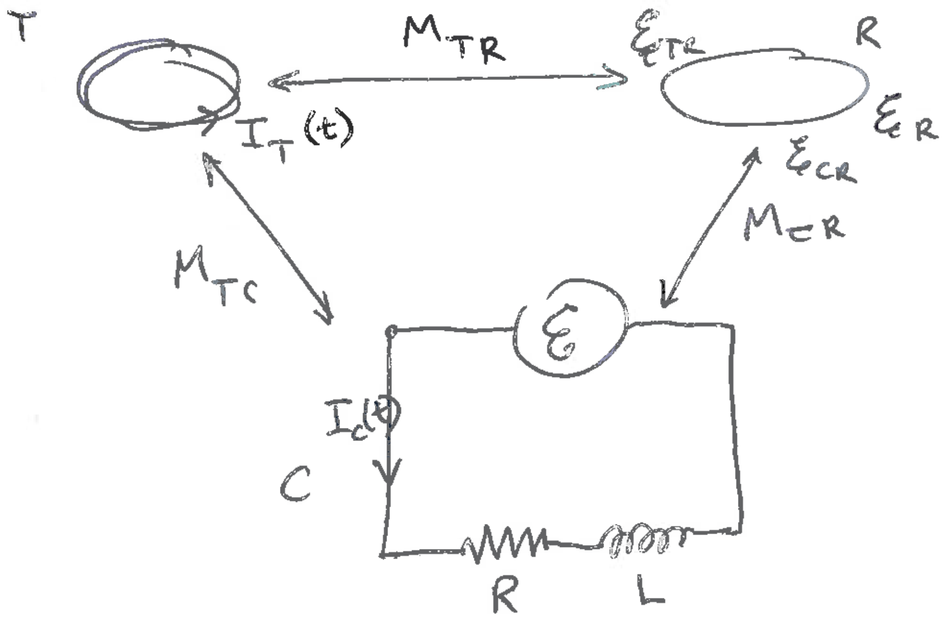
$$\sigma = \rho^{-1}$$

$S\,m^{-1} = [\sigma]$ $\quad \Omega\,m = [\rho]$

mho $\mho = \Omega^{-1}$ ohm

$$-\frac{d\Phi}{dt} = \mathcal{E}$$





$$\mathcal{E} - V_L - V_R = 0$$

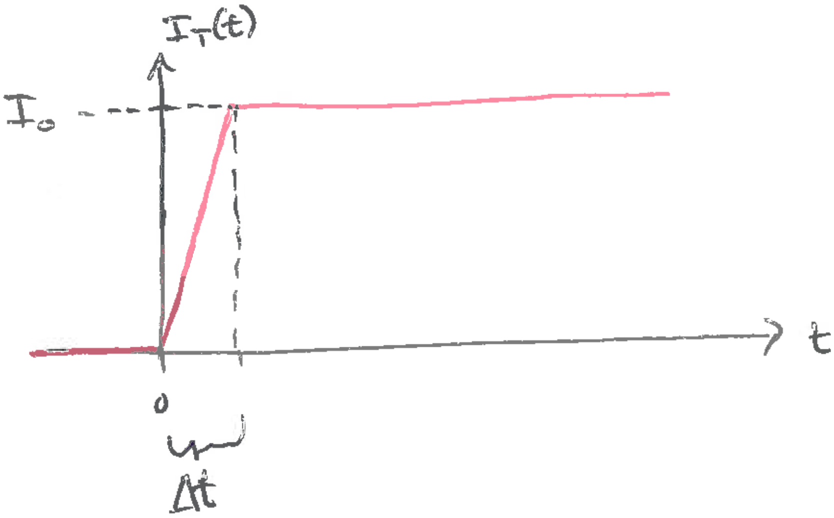
$$\mathcal{E} = -M_{TC} \frac{dI_T(t)}{dt}$$

$$V_L = L \frac{dI_c(t)}{dt}$$ RETARDING

$$V_R = R I_c(t)$$ DISSIPATION

TRANSMITTER CURRENT, UNDER OUR CONTROL !!

I_T



$$I_T(t) = I_o\, u(t)$$

↑ RAMP FUNCTION

$$u(t) = \begin{cases} 0 & t < 0 \\ t/\Delta t & 0 \le t \le \Delta t \\ 1 & t > \Delta t \end{cases}$$

$$\mathcal{E} = -M_{Tc} \frac{dI(t)}{dt}$$

$$= -M_{Tc} I_o \frac{du(t)}{dt}$$

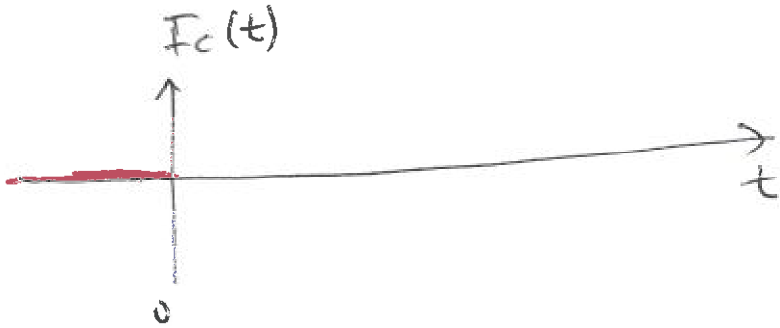
$$-M_{Tc} I_o \frac{du(t)}{dt} = L \frac{dI_c(t)}{dt} + R I_c(t) \qquad (\star)$$

$$\frac{du(t)}{dt} = \begin{cases} \frac{d0}{dt} = 0 & t < 0 \\ \frac{dt}{\Delta t\, dt} = \frac{1}{\Delta t} & 0 \le t \le \Delta t \\ \frac{d1}{dt} = 0 & t \geq \Delta t \end{cases}$$

1: $t < 0$, $\frac{du(t)}{dt} = 0$

$$0 = L \frac{dI_c(t)}{dt} + RI_c(t)$$

$I_c(t) = 0 \qquad t < 0$



2: $0 \leq t \leq \Delta t$, $\frac{du(t)}{dt} = \frac{1}{\Delta t}$

$$-\frac{M_{TC} I_c}{\Delta t} = L \frac{dI_c(t)}{dt} + RI_c(t)$$

OF THE FORM: $C_2 = C_0 \frac{dy}{dt} + C_1 y$

$$y = y_0 \exp\left(-t \frac{C_1}{C_0}\right) + \frac{C_2}{C_1}$$

$y(0)$, $t = 0$ INITIAL CONDITION

$$y(0) = 0 = y_0 \cancelto{1}{\exp\left(-0 \frac{C_1}{C_0}\right)} + \frac{C_2}{C_1}$$

$$0 = y_0 + \frac{C_2}{C_1}$$

$$-\frac{C_2}{C_1} = y_0$$

$$I_c(t) = -\frac{M_{TC} I_0}{R \Delta t}\left[1 - \exp\left(-t \frac{R}{L}\right)\right]$$

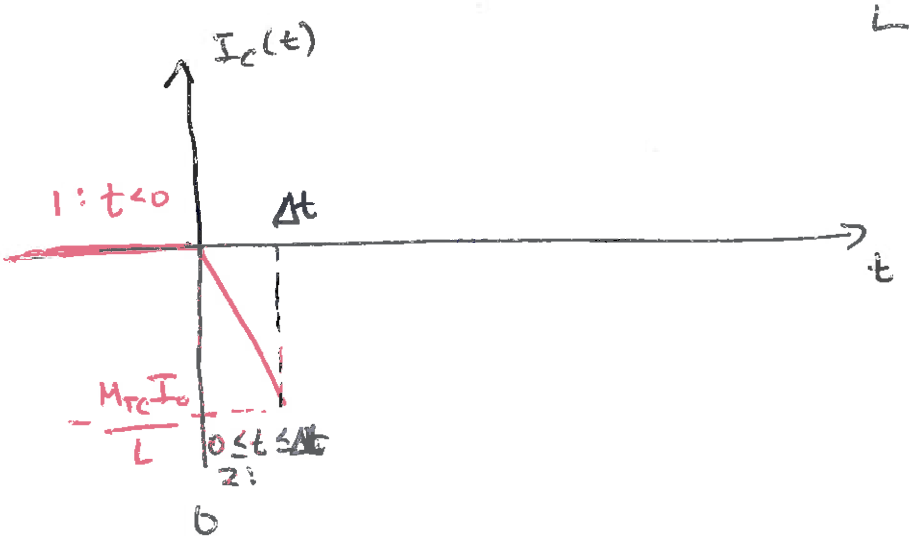
$\boxed{\text{ASSUMPTION: } t \ll \frac{L}{R}}$

$$\exp(x) = 1 - x + \frac{1}{2}x^2 - \ldots \qquad \text{TAYLOR SERIES}$$

$$I_c(t) = -\frac{M_{Tc} I_o}{R\Delta t}\left[1 - \left(1 - \left(-t\frac{R}{L}\right)\right)\right]$$

$$I_c(t) = -\frac{M_{Tc} I_o}{L\Delta t}\, t$$

IF $t = \Delta t$ $\quad I_c(t) = -\dfrac{M_{Tc} I_o}{L}$



$3:\ t > \Delta t, \quad \dfrac{du(t)}{dt} = 0$

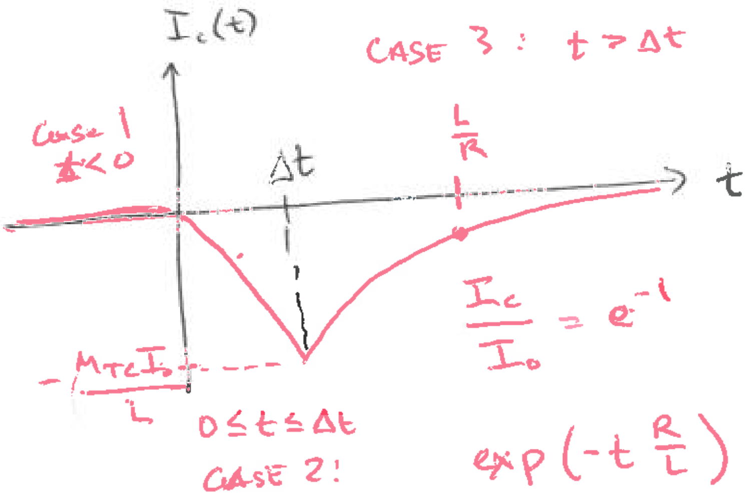
$$0 = L\frac{dI_c(t)}{dt} + R\, I_c(t)$$

$$0 = a_o\frac{dy}{dt} + a_1 y$$

$$y = k\exp\left(-t\frac{a_1}{a_o}\right)$$

$$y(0) = I_c(\Delta t) = -\frac{M_{Tc} I_o}{L}$$

$$I_c(t) = -\frac{M_{Tc} I_o}{L}\exp\left(-t\frac{R}{L}\right), \quad t > \Delta t$$



$$\exp\left(-t \frac{R}{L}\right)$$

$$t \frac{R}{L} = 1$$

$$t = \frac{L}{R}$$

$R >> L$, RESISTIVE EARTH = RAPID DECAY

$R << L$, CONDUCTIVE EARTH = SLOW DECAY